MIL-H-42067
15 June 1989

MILITARY SPECIFICATION

# HUMIDIFIER, OPERATING ROOM

This specification is approved for use by all departments and agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of humidifier for use in the operating rooms of the Medical Unit - Self Contained Transportable (MUST).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 4520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-636 | Box, Shipping, Fiberboard |

Military

| | |
|---|---|
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible |
| MIL-M-7298 | Manuals, Commercial Off-The-Shelf |
| MIL-P-116 | Preservation, Methods of |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series: General Specification for |
| MIL-T-704 | Treatment and Painting of Material |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U.S. Military Property |
| MIL-STD-147 | Palletized Unit Loads |
| MIL-STD-794 | Parts and Equipment, Procedures for Packaging of |

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

DRAWING

Defense Personnel Support Center (DPSC)

24346 Humidifier, Shelter, Surgery (MUST)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are issues of the documents cited in the solicitation.

Underwriter's Laboratories, Inc. (UL)

UL-73 - Electric Motor-Operated Appliance
UL-507 - Electric Fans

(Application for copies should be addressed to the Underwriters' Laboratories, Inc. 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, Long Island, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 05050).

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Fire and casualty hazards.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

3.2.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL 507. Proof shall consist of one of the following:

a. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

b. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government, which reserves the right to require a facility's report and to conduct a facility survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified.

3.2.2 Compliance. Compliance with above paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.3 Material. Materials shall be in accordance with DPSC drawing No. 24346 and as hereinafter specified. All metals shall be corrosion-resisting or treated to resist corrosion. Materials not specified shall be of the highest quality and suitable in all respects for their intended purpose.

3.4 Design and construction. The design and construction of the humidifier shall be in accordance with DPSC Drawing No. 24346 and as hereinafter specified. The humidifier shall consist of a housing containing a water reservoir and an air circulating fan, and the other components specified on the drawing. Water level shall be maintained by a float valve in the reservoir. The water in the reservoir shall be heated by the electric heaters in the reservoir. The fan shall draw air across the evaporating water in the reservoir, and discharge the resultant humid air into the operating room of the MUST unit. A drain and valve shall be provided to permit draining of the reservoir. A humidity sensing control valve shall be provided to regulate the operation of the unit about a selectable humidity setting.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

3.4.1 Box assembly. The box assembly shall be fabricated of 6061-T4 and T6 aluminum alloy and shall have outer dimensions, excluding mounting plates, of 36 inches X 12 inches X 8 inches. As an alternate to riveting of the box assembly, spot welding may be used. When spotwelding is used, the appropriate holes shall be eliminated from the parts. Spotwelds shall be placed at each rivet location as minimum.

3.4.2 Pan assembly. The pan assembly (water reservoir) shall be fabricated of 6061-T4 aluminum alloy and shall have outer dimensions of 20 inches X 6.75 inches X 3.00 inches. The fabricated pan shall be leakproof.

3.4.3 Deflector assembly. The deflector assembly shall be fabricated of 6061-T4 aluminum alloy per DPSC Drawing 24346.

3.5 Electrical power and water inputs.

3.5.1 Electrical connector. The power input connector shall be an integral part of the unit wiring harness and shall be accessible from the rear of the humidifier. The connector shall mate with the receptable on the MUST expandable shelter wall. The cable configuration shall conform to DPSC Drawing No. 24346.

3.5.2 Electrical power requirements. The humidifier shall operate on 120 volt, single and three phase, 400 Hz power.

3.5.2.1 Heating elements. The humidifier shall be equipped with three 600 watt heating elements requiring 120 volts, 3 phase, 400 Hz power.

3.5.2.2 Blower fan. The flower fan shall have a capability of 120 CFM and shall operate on 120 volt, single phase, 400 Hz power.

3.5.3 Water connection. A water inlet and drain port shall be provided on the side of the humidifier. The drain port shall be controlled by a two-position tank drain valve on the front of the humidifier. The water hoses, hose support clamps, and fittings required to connect the humidifier to the shelter, water and provisions shall be furnished.

3.6 Humidistat control. The humidistat will be located on the opposite wall from the humidifier in the shelter. The humidistat shall be capable of controlling the relative humidity to within plus or minus 10 percent of the set point over a range of 20 to 80 percent. Control wiring for connecting the humidistat to the humidifier is provided within the MUST expandable shelter wiring system. A mounting plate to permit mounting of the control in the shelter shall be provided.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

3.7 Electrical design.

3.7.1 Unit performance. The humidifier shall operate as required when supplied with electrical power specified in 3.5.2.

3.7.2 Thermal concept. The unit shall be built with use of the best thermal design. A thermostat shall operate to remove power from the heater elements in the event of an overheating condition or lack of water in the tank.

3.7.3 Unmated connectors. Unmated connectors shall be protected from contamination that might adversely affect the electrical conducting properties of the contacts. Captive-type dust caps shall be provided where necessary and attached to the equipment.

3.7.4 Stress relief. Whenever a possible displacement exists between the body of a part and its lead terminations, the leads shall be formed to furnish stress relief.

3.7.5 Electrical grounding - Bonding. The unit shall be constructed so that all common returns are joined at a common point. The case ground shall also be terminated at the specified connector pin.

3.8 General requirements.

3.8.1 Parts marking. Each part and assembly shall be marked with its part number, preceded by the DPSC drawing number. Exceptions to the above include (1) parts that do not have suitable or sufficient surface area for the number, and (2) parts that are permanently attached to an overall assembly by welding, brazing, soldering, or riveting.

3.8.2 Screw threads. All conventional straight screw threads shall conform to MIL-S-7742.

3.8.3 Spotwelding. Prior to spotwelding, all parts to be spotwelded shall be thoroughly cleaned and etched. The spotwelding apparatus shall be suitable for the gages and types of materials being spotwelded. The spotwelds shall completely fuse the materials being welded and shall be stronger than the base metals.

3.9 Identification. The unit shall be identified with black stencil paint in conformance with FED-STD-595, color chip No. 17038. The paint used shall meet all conventional conditions described herein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

3.10 Finish. The entire unit shall be finished to prevent corrosion and provide a pleasing appearance.

3.10.1 Housing. The aluminum housing shall be finished on all surface per MIL-T-704, Type B. The color of the finish enamel shall be No. 24516 of FED-STD-595.

3.10.2 Pan assembly. The exterior of the pan assembly shall be finished in accordance with MIL-T-704, Type B. The color of the finish enamel shall be No. 24516 of FED-STD-595.

3.10.3 Hardware. Hardware shall be appropriately plated to resist corrosion.

3.11 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. Wiring diagrams schematics shall be provided for electrical components. All parts, except common hardware, shall be identified in the parts list by both the manufacturer's and the contractor's part numbers. As soon as practicable after the award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract, or purchase order. In addition, the contractor shall furnish 10 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission copies shall be at the option of the government.

3.12 Workmanship. The unit shall be finished according to the best construction practices for high-quality electrical equipment. Particular attention shall be given to neatness and thoroughness of soldering, wire routing, impregnation of coated parts, part marking, welding and brazing, lacquering, and plating. In addition, electrical clearance between components and terminals, foreign matter, rugged construction, and component support or mountings shall be a major consideration.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.3 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. Sampling for examination.

| | Inspection level | AQL (defects per hundred units) |
|---|---|---|
| For visual examination | | |
| Major defects | S-4 | 1.0 |
| Minor defects | S-4 | 4.0 |
| For dimensional examination | S-1 | 2.5 |

4.2.2 For tests. Sampling for tests shall be 100 percent, except for first article test where only one unit shall be tested.

4.3 Examination. The units shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in table II.

TABLE II. Categories and Defects.*

| Categories | Defects |
|---|---|
| Major | |
| 101 | Any component broken, missing, or otherwise unserviceable. |
| 102 | Unit not free of burrs and sharp edges on exposed accessible surfaces. |
| 103 | Leaks in water reservoir. |
| 104 | Unit not properly grounded. |
| Minor | |
| 201 | Service data not enclosed with unit. |
| 202 | Finish not free of scratches and nicks. |
| 203 | Identification marking not as specified. |
| 204 | Unit not clean, not free of grease, oil, or foreign matter. |

* Examination shall not be limited to the classified possible defects listed herein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

4.3.2 Dimensional examination. The units shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Operation test. The unit shall be connected to a 3 phase, 120 volt, 400 Hertz power source without water in the pan. The unit shall be examined to assure that the fan operates, the heaters operate and that the thermostat de-energizes the immersion heaters. Immediately following this, the water input line shall be connected to a source of water and the pan permitted to fill. The water connections shall be observed for leakage and the float valve shall be observed for proper adjustment and operation. The water connection shall be removed and the drain valve opened. The pan and drain plumbing shall be observed for leaks.

4.4.2 Humidity control test. The humidity control shall be attached to a level vertical wall in accordance with the manufacturer's instructions. The relative humidity of the air in the test room shall be maintained between 50 and 60 percent. The accuracy of the humidity control shall be determined and adjustments made as necessary. In lieu of the actual test, a certificate from the manufacturer of the controls stating that it will maintain an accuracy of plus or minus ten (10) percent over the range of 20 to 80 percent relative humidity shall be acceptable.

4.4.3 First article test. The first article sample shall be mounted in a "MUST" surgical shelter to ascertain its ability to interface as intended. The unit shall be tested as specified in 4.4.1 and 4.4.2. This test may be waived in the case of a contractor who has performed it successfully under contract within five (5) years prior to the date of the current contract or solicitation.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each humidifier shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each humidifier shall be packaged in accordance with MIL-P-116, method III in a box conforming to PPP-B-636, class domestic. Humidifier shall be cushioned within the box to prevent shifting and damage to contents or the container. Closure shall be as specified in the box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for humidifier shall conform to the requirements of MIL-STD-794 for Level C.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Humidifier packaged as specified in 5.1.1.3 shall require no further packing except that the box shall conform to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the box specification.

5.2.2 Level B. Humidifier packaged as specified in 5.1.1.3 shall require no further packing.

5.2.3 Level C. The level C packing for humidifier shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in length, 52 inches in width, and 54 inches in height. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Humidifiers packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Humidifiers packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

6.2 Aquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. National Stock Number (NSN)
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2)
d. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

6.3 This specification covers the requirements for the following item.

| NSN | Item Identification |
|---|---|
| 4520-01-071-1773 | HUMIDIFIER, Operating Room, 120 Volt, Single and Three Phase, 400 Hz, AC |

6.4 Subject term (key word) listing.

Shelter
Surgical
Operating Room
Surgery
MUST

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DLA - DM

Agent:

DCSC-CS

Civil Agency Coordinating Activities:

VA
FDA

Project No. 4520-0304

Location:
MILSP/HUMIDIFIER/H13-34



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:19Z
Check the source to verify that this is the current version before use.